

AURA

Academic Unified Response & Assistance

Software Engineering Project

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Milestone - 5
Report

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APIs Tests

Test Case 1.1

Endpoint	/api/auth/signup
Method	POST
Test Case	Register a new user account with email, password, full name, and role.

Input:

```
{
  "email": "admin_test@example.com",
  "password": "AdminPass123!",
  "full_name": "Admin User",
  "role": "admin"
}
```

Expected Output:

```
{
  "status_code": 201,
  "body": {
    "access_token": "<JWT_TOKEN>",
    "refresh_token": "<REFRESH_TOKEN>",
    "token_type": "bearer",
    "expires_in": 3600,
    "user": {
      "email": "admin_test@example.com",
      "role": "admin",
      "full_name": "Admin User",
      "is_active": true
    }
  }
}
```

Actual Output:

```
{
  "status_code": 201,
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "refresh_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "bearer",
  "expires_in": 3600,
  "user": {
    "id": 10,
    "email": "admin_test@example.com",
    "role": "admin",
    "full_name": "Admin User",
    "is_active": true,
    "created_at": "2025-11-30T16:05:09"
  }
}
```

APIs Tests

```
}
```

APIs Tests

Test Case 1.2

Endpoint	/api/auth/login
Method	POST
Test Case	Authenticate existing user and return JWT tokens with user profile.

Input:

```
{
  "email": "student@test.com",
  "password": "StudPass123!"
}
```

Expected Output:

```
{
  "status_code": 200,
  "body": {
    "access_token": "<JWT_TOKEN>",
    "refresh_token": "<REFRESH_TOKEN>",
    "token_type": "bearer",
    "expires_in": 3600,
    "user": {
      "email": "student@test.com",
      "role": "student"
    }
  }
}
```

Actual Output:

```
{
  "status_code": 200,
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "refresh_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "bearer",
  "expires_in": 3600,
  "user": {
    "id": 11,
    "email": "student@test.com",
    "role": "student",
    "full_name": "Student User",
    "is_active": true,
    "created_at": "2025-11-30T16:05:09"
  }
}
```

APIs Tests

Test Case 2.1

Endpoint	/api/quizzes/generate
Method	POST
Test Case	Generate quiz questions using Google Gemini LLM based on topic, difficulty, and course context.

Input:

```
{
  "topic": "Python Data Structures",
  "num_questions": 5,
  "difficulty": "medium",
  "course_id": 1
}
```

Expected Output:

```
{
  "status_code": 201,
  "body": {
    "id": 15,
    "title": "Python Data Structures Quiz",
    "topic": "Python Data Structures",
    "difficulty": "medium",
    "questions": [
      {
        "text": "What is a list in Python?",
        "options": ["A mutable sequence", "An immutable sequence", "A dictionary", "A set"],
        "correct_answer": "A mutable sequence",
        "explanation": "Lists are mutable ordered sequences"
      }
    ],
    "total_marks": 10,
    "is_published": false
  }
}
```

Actual Output:

```
{
  "status_code": 422,
  "detail": [
    {
      "type": "missing",
      "loc": ["body", "use_latex"],
      "msg": "Field required"
    }
  ]
}
```

APIs Tests

```
}
```

APIs Tests

Test Case 2.2

Endpoint	/api/quizzes/1/submit
Method	POST
Test Case	Submit student answers and receive AI-powered automated grading.

Input:

```
{
  "answers": [
    {"question_index": 0, "selected_options": ["A mutable sequence"]},
    {"question_index": 1, "selected_options": ["str", "int", "list"]}
  ]
}
```

Expected Output:

```
{
  "status_code": 200,
  "body": {
    "id": 1,
    "quiz_id": 1,
    "user_id": 5,
    "score": 8,
    "total_marks": 10,
    "percentage": 80,
    "submitted_answers": [...]
  }
}
```

Actual Output:

```
{
  "status_code": 404,
  "detail": "Quiz not found or not published"
}
```


APIs Tests

Test Case 3.1

Endpoint	/api/slide-decks/generate
Method	POST
Test Case	Generate educational slide deck with AI-powered content and visualizations using Google Gemini.

Input:

```
{
  "course_id": 1,
  "title": "Introduction to Python",
  "description": "Comprehensive Python basics",
  "topics": ["Variables", "Data Types", "Control Flow"],
  "num_slides": 5,
  "format": "presentation"
}
```

Expected Output:

```
{
  "status_code": 201,
  "body": {
    "id": 10,
    "title": "Introduction to Python",
    "slides": [
      {
        "title": "Python Variables",
        "content": "# Variables\n- Store data values\n- Dynamic typing",
        "graph_data": {
          "type": "bar",
          "title": "Variable Types Usage",
          "labels": ["int", "str", "list"],
          "datasets": [{"label": "Frequency", "data": [40, 35, 25]}]
        }
      }
    ],
    "course_id": 1,
    "created_by_id": 3
  }
}
```

Actual Output:

```
{
  "status_code": 201,
  "id": 10,
  "title": "Introduction to Python",
  "description": "Comprehensive Python basics",
  "slides": [
```

APIs Tests

```
{
  "title": "Introduction to Python",
  "content": "# Welcome to Python\n\n- Versatile language\n- Web to data science",
  "graph_data": {
    "type": "bar",
    "title": "Language Popularity",
    "labels": ["Python", "JavaScript", "Java"],
    "datasets": [{"label": "Usage %", "data": [45, 30, 25]}]
  },
  "graph_image": null
},
{
  "title": "Variables and Data Types",
  "content": "## Common Types\n\n- **Integers**: `x = 10`\n- **Strings**: `name = \"AURA\"`,
  "graph_data": null,
  "graph_image": null
}
],
"course_id": 1,
"created_by_id": 3,
"created_at": "2025-11-30T16:10:15"
}
```

Test Case 3.2

Endpoint	/api/slide-decks/preview
Method	POST
Test Case	Generate preview outline before full slide generation (faster, cheaper LLM call).

Input:

```
{
  "course_id": 1,
  "title": "Advanced Algorithms",
  "topics": ["Sorting", "Graph Algorithms", "Dynamic Programming"],
  "num_slides": 8,
  "format": "presentation"
}
```

Expected Output:

```
{
  "status_code": 200,
  "outline": [
    "Slide 1: Introduction to Algorithms - Overview and importance",
    "Slide 2: Sorting Algorithms - Bubble, Quick, Merge sort comparison",
    "Slide 3: Graph Basics - Nodes, edges, representations",
    "Slide 4: BFS and DFS - Traversal algorithms",
    "Slide 5: Shortest Path - Dijkstra and Bellman-Ford",
    "Slide 6: Dynamic Programming Intro - Memoization vs tabulation",
    "Slide 7: Classic DP Problems - Knapsack, LCS examples",
    "Slide 8: Summary - Key takeaways and practice resources"
  ]
}
```

Actual Output:

```
{
  "status_code": 200,
  "outline": [
    "Slide 1: Introduction to Algorithms - definition and significance in computer science",
    "Slide 2: Sorting Algorithms - overview of common techniques (bubble, insertion, quick, merge)",
    "Slide 3: Complexity Analysis - Big O notation for sorting algorithms",
    "Slide 4: Graph Basics - graph terminology and representations",
    "Slide 5: Graph Traversal - BFS and DFS algorithms with examples",
    "Slide 6: Dynamic Programming Fundamentals - overlapping subproblems and optimal substructure",
    "Slide 7: DP Example Problems - Fibonacci, knapsack problem",
    "Slide 8: Conclusion - summary and further learning resources"
  ]
}
```

APIs Tests

```
}
```

APIs Tests

Test Case 4.1

Endpoint	/api/chatbot/chat
Method	POST
Test Case	Verify chatbot requires authentication and rejects unauthorized requests.

Input:

```
{
  "message": "Hello, explain Python decorators",
  "mode": "academic"
}
```

Expected Output:

```
{
  "status_code": 401,
  "detail": "Not authenticated"
}
```

Actual Output:

```
{
  "status_code": 401,
  "detail": "Not authenticated"
}
```

APIs Tests

Test Case 4.2

Endpoint	/api/chatbot/chat
Method	POST
Test Case	Conduct academic conversation with AI chatbot using Google ADK with memory and observability.

Input:

```
{
  "message": "Explain the difference between lists and tuples in Python",
  "mode": "academic",
  "conversation_id": "conv-123"
}
Headers: Authorization: Bearer <TOKEN>
```

Expected Output:

```
{
  "status_code": 200,
  "response": "Lists and tuples are both sequence types in Python:\n\n**Lists:**\n- Mutable (can be modified)\n- Syntax: [1, 2, 3]\n- Methods: append(), remove(), etc.\n\n**Tuples:**\n- Immutable (cannot be modified)\n- Syntax: (1, 2, 3)\n- Faster, memory-efficient\n- Use for fixed collections",
  "conversation_id": "conv-123",
  "mode": "academic"
}
```

Actual Output:

```
{
  "status_code": 200,
  "response": "***Lists vs Tuples in Python:**\n\n**Lists** are mutable sequences defined with square brackets []. You can add, remove, or modify elements after creation using methods like append(), remove(), and pop(). Example: `my_list = [1, 2, 3]`\n\n**Tuples** are immutable sequences defined with parentheses (). Once created, you cannot change their contents. They're faster and use less memory. Example: `my_tuple = (1, 2, 3)`\n\n**Key Difference:** Mutability. Use lists when you need to modify data, tuples for fixed collections or as dictionary keys.",
  "conversation_id": "conv-123",
  "mode": "academic",
  "tokens_used": 156
}
```

APIs Tests

Test Case 4.3

Endpoint	/api/chatbot/chat
Method	POST
Test Case	Chatbot uses integrated DuckDuckGo search tool for current information.

Input:

```
{
  "message": "What are the latest Python 3.12 features?",
  "mode": "general",
  "use_web_search": true
}
```

Expected Output:

```
{
  "status_code": 200,
  "response": "Based on web search results, Python 3.12 introduces:\n\n1. **Per-Interpreter GIL** - Better multi-threading\n2. **Type Parameter Syntax** - PEP 695 generic syntax\n3. **f-string improvements** - More flexible formatting\n4. **Performance gains** - 5% faster than 3.11\n\nSources: [python.org, realpython.com]",
  "sources_used": ["web_search"],
  "conversation_id": "conv-456"
}
```

Actual Output:

```
{
  "status_code": 200,
  "response": "Python 3.12 features several improvements:\n\n1. **Per-Interpreter GIL** - Each interpreter has its own Global Interpreter Lock, enabling better parallelism\n2. **Improved f-strings** - More expressive string formatting with nested expressions\n3. **Type Parameter Syntax** (PEP 695) - Cleaner generic type hints: `def func[T](x: T) -> T`\n4. **Performance** - Up to 5% faster than Python 3.11\n5. **Better error messages** - More helpful traceback information\n\nThese changes make Python faster and more developer-friendly.",
  "sources_used": ["web_search"],
  "search_queries": ["Python 3.12 new features"],
  "conversation_id": "conv-456",
  "tokens_used": 203
}
```

APIs Tests

Test Case 5.1

Endpoint	/api/queries/
Method	POST
Test Case	Attempt to create query with missing required fields or invalid enum values.

Input:

```
{
  "title": "Need help with REST design",
  "description": "What are best practices?",
  "category": "CONCEPTUAL",
  "priority": "MEDIUM"
}
```

Expected Output:

```
{
  "status_code": 201,
  "body": {
    "id": 5,
    "title": "Need help with REST design",
    "status": "OPEN",
    "category": "CONCEPTUAL",
    "priority": "MEDIUM",
    "student_id": 11
  }
}
```

Actual Output:

```
{
  "status_code": 422,
  "detail": [
    {
      "type": "missing",
      "loc": ["body", "student_id"],
      "msg": "Field required",
      "input": {
        "title": "Need help with REST design",
        "description": "What are best practices?",
        "category": "CONCEPTUAL",
        "priority": "MEDIUM"
      }
    }
  ]
}
```


APIs Tests

Test Case 5.2

Endpoint	/api/analytics/overview
Method	GET
Test Case	Student attempts to access admin-only analytics endpoint.

Input:

```
Method: GET
Headers: Authorization: Bearer <STUDENT_TOKEN>
```

Expected Output:

```
{
  "status_code": 403,
  "detail": "Admin access required"
}
```

Actual Output:

```
{
  "status_code": 403,
  "detail": "Admin access required"
}
```